MINI RESEARCH PROJECT · PHARMACEUTICAL BIOTECHNOLOGY

Artificial Intelligence in *Drug Discovery* & Repurposing

How machine learning, deep neural networks, and generative AI are reshaping the pharmaceutical pipeline — from target identification to clinical repurposing.

SUBJECT

Role of AI & Biotechnology in Pharma

TOPIC

AI in Drug Discovery & Repurposing

FORMAT

Mini Research Project (5–6 Pages)

YEAR

2025

NAVIGATION

Table of Contents

SECTION 01

01 Abstract

The pharmaceutical industry is undergoing a fundamental transformation driven by artificial intelligence (AI). Traditional drug development — historically characterised by timelines of 10–15 years and costs exceeding US $2.6 billion per approved molecule — is being accelerated through machine learning (ML), deep learning, and generative models that can analyse vast biomedical datasets, predict molecular behaviour, and identify new therapeutic uses for existing compounds. This paper examines the current applications of AI across the drug discovery and repurposing continuum, reviews landmark case studies including the COVID-19 pandemic response, and critically evaluates the technological, regulatory, and ethical challenges that accompany this paradigm shift.

Keywords: Artificial intelligence, drug discovery, drug repurposing, machine learning, deep learning, molecular docking, virtual screening, generative chemistry.

SECTION 02

02 Introduction

Drug discovery is among the most intellectually demanding and financially intensive endeavours in modern science. Despite decades of technological progress, the probability of a candidate molecule successfully advancing from preclinical research to regulatory approval remains below 10% [1]. The resulting inefficiency imposes enormous costs on healthcare systems and delays access to life-saving therapies for patients worldwide.

Artificial intelligence — and in particular machine learning sub-disciplines such as deep neural networks, reinforcement learning, and large language models — has emerged as a transformative force that addresses several of these inefficiencies simultaneously. By learning complex, nonlinear patterns from large-scale biological and chemical datasets, AI systems can prioritise drug targets, predict toxicity, optimise molecular scaffolds, and identify candidate compounds orders of magnitude faster than traditional high-throughput screening [2].

> KEY DEFINITION
>
> **Drug Repurposing** (also called drug repositioning or re-profiling) refers to the identification of new therapeutic indications for approved or investigational drugs, thereby bypassing early-stage safety studies and significantly compressing development timelines.

Drug repurposing is a particularly fertile application domain for AI. Approved drugs already carry well-characterised safety profiles, meaning that a successful repurposing candidate can enter Phase II clinical trials almost immediately, reducing average development costs by up to 60%

compared with de novo discovery [3]. The global AI in drug discovery market was valued at approximately US $1.7 billion in 2023 and is projected to exceed US $9.8 billion by 2030, reflecting the sector's rapid adoption of these technologies [4].

$0B

AVERAGE COST PER APPROVED DRUG (TRADITIONAL)

0 yrs

AVERAGE DEVELOPMENT TIMELINE WITHOUT AI

0%

POTENTIAL COST REDUCTION VIA REPURPOSING

SECTION 03

03 The Traditional Drug Discovery Pipeline & Its Limitations

The canonical drug development pipeline proceeds through several well-defined stages: target identification, hit discovery via compound library screening, lead optimisation, preclinical animal studies, and multi-phase clinical trials before regulatory submission. Each stage is characterised by attrition — a compounding series of failures driven primarily by insufficient efficacy, unacceptable toxicity, and unfavourable pharmacokinetics [1].

STAGE 1	→	STAGE 2	→	STAGE 3	→	STAGE 4	→	STAGE 5	→	STAGE 6
Target ID & Validation AI		**Hit Discovery** AI		**Lead Optimisation** AI		**Preclinical Studies**		**Phase I–III Trials**		**Regulatory Approval**

Stages highlighted in green represent primary AI intervention points.

High-throughput screening (HTS) — the workhorse of conventional hit discovery — can test millions of compounds but generates enormous false-positive rates and remains blind to multi-target pharmacology [5]. Lead optimisation, which seeks to improve potency, selectivity, and pharmacokinetic properties simultaneously, traditionally relies on the intuition of medicinal chemists and iterative synthesis cycles that can span years. These inefficiencies have contributed to a phenomenon termed "Eroom's Law" — the observation that drug development productivity has halved roughly every nine years since the 1950s [6].

SECTION 04

04 AI Technologies in Drug Discovery

4.1 Machine Learning & Deep Neural Networks

Supervised ML models — including random forests, gradient boosted trees, and graph neural networks (GNNs) — have demonstrated state-of-the-art performance in quantitative structure–activity relationship (QSAR) modelling, predicting how chemical modifications affect biological activity [7]. GNNs in particular exploit the natural graph-like topology of molecules, treating atoms as nodes and bonds as edges, enabling nuanced representation learning that outperforms traditional fingerprint-based descriptors [8].

Recurrent neural networks and transformers have been applied to protein sequence data, learning evolutionary and structural features that inform target druggability assessment. The protein language model ESM-2, developed by Meta AI, encodes structural and functional properties from sequence alone, dramatically accelerating target validation workflows [9].

4.2 AlphaFold & Structural Biology

Perhaps the single most consequential AI breakthrough for drug discovery is DeepMind's AlphaFold2, which achieved near-experimental accuracy in predicting three-dimensional protein structures from amino acid sequences [10]. Prior to AlphaFold, experimental structure determination via X-ray crystallography or cryo-EM could take months to years. AlphaFold's release of over 200 million predicted structures has unlocked structure-based drug design for entire classes of previously "undruggable" targets, including intrinsically disordered proteins implicated in cancer and neurodegeneration [11].

> "*AlphaFold has already been used to unlock new avenues of research across biology and medicine — from designing drugs to understanding diseases.*"
>
> *— NATURE, 2022 [10]*

4.3 Generative AI & de Novo Molecular Design

Generative models — including variational autoencoders (VAEs), generative adversarial networks (GANs), and transformer-based architectures such as Insilico Medicine's REINVENT — learn the latent chemical space of known drug-like molecules and generate novel structures with desired properties [12]. These models can be conditioned on specific target-binding poses, selectivity profiles, synthetic accessibility scores, or ADMET (absorption, distribution, metabolism, excretion, toxicity) constraints, effectively allowing chemists to navigate molecular space guided by multiparameter objectives [13].

05 AI in Drug Repurposing

5.1 Network-Based Approaches

Biological systems can be represented as networks (or graphs) in which nodes represent genes, proteins, or diseases, and edges encode experimentally determined interactions. AI algorithms applied to these networks — collectively termed network medicine — can identify "drug–disease proximity": the graph-theoretic distance between a drug's targets and the molecular modules dysregulated in a given disease [14]. Drugs demonstrating high proximity to a new disease signature become repurposing candidates, even if their original indication is entirely unrelated.

5.2 Knowledge Graph Embeddings

Knowledge graphs (KGs) integrate heterogeneous biomedical data — clinical records, omics data, literature, drug–target interactions, and adverse event reports — into a unified relational structure. Embedding algorithms such as TransE and RotatE learn low-dimensional vector representations of entities and relations, enabling link prediction that surfaces novel drug–disease associations [15]. Platforms such as BenevolentAI and Healx rely on proprietary KG systems to generate repurposing hypotheses at scale, several of which have advanced into clinical trials.

5.3 Natural Language Processing on Biomedical Literature

PubMed currently indexes over 35 million biomedical citations, a corpus too large for manual review by any research team. NLP pipelines — including named entity recognition (NER), relation extraction, and BERT-derived biomedical language models such as BioBERT and PubMedBERT — automatically mine drug–target–disease associations from unstructured text [16]. Such approaches frequently surface repurposing signals hidden within obscure mechanistic side-notes years before they enter mainstream awareness.

AI METHOD	PRIMARY DATA SOURCE	KEY ADVANTAGE	NOTABLE PLATFORM
Graph Neural Networks	Molecular graphs, PPI networks	Captures structural topology	Atomwise, Schrödinger
Knowledge Graph Embeddings	Multi-omics, clinical data	Multi-modal integration	BenevolentAI, Healx
NLP / Language Models	PubMed, clinical notes	Mines unstructured literature	IBM Watson Health
Generative Models (VAE/GAN)	Chemical libraries	De novo molecule generation	Insilico Medicine
Reinforcement Learning	Simulated docking environments	Multi-objective optimisation	Exscientia

06 Case Studies

6.1 COVID-19 Pandemic Response

The SARS-CoV-2 pandemic provided an unprecedented stress test for AI-driven drug repurposing. Within weeks of the virus's structure being published, platforms such as BenevolentAI used their KG infrastructure to identify baricitinib — approved for rheumatoid arthritis — as a candidate capable of inhibiting AAK1-mediated viral endocytosis and moderating the cytokine storm associated with severe COVID-19 [17]. This hypothesis, generated in under 48 hours, was subsequently validated in the ACTT-2 randomised trial and the drug received emergency use authorisation from the FDA in 2020 [17].

Concurrently, MIT's AI Cures project and Deargen's MT-DTI model both predicted atazanavir and remdesivir as high-priority candidates based on drug–target interaction models trained on existing antiviral data [18]. Remdesivir's subsequent authorisation demonstrated that AI-ranked repurposing hypotheses can reach clinical validation within months under emergency conditions.

6.2 Insilico Medicine: First AI-Designed Drug in Clinical Trials

In 2023, Insilico Medicine announced that ISM001-055, a drug designed entirely by its generative AI platform for the treatment of idiopathic pulmonary fibrosis (IPF), had entered Phase II clinical trials [12]. The discovery phase, which would conventionally take 4–5 years, was compressed to 18 months using a pipeline integrating PandaOmics (target identification) and Chemistry42 (molecular generation). This milestone represents the first fully AI-conceived drug candidate to reach late-stage human trials.

6.3 Exscientia & Atomwise: Accelerating Lead Discovery

Exscientia, in collaboration with Sumitomo Dainippon Pharma, developed DSP-1181 — an obsessive-compulsive disorder (OCD) candidate — in approximately 12 months versus an industry average of 4.5 years, using reinforcement learning-guided molecular optimisation [19]. Similarly, Atomwise's AtomNet platform identified candidate inhibitors for the Ebola virus VP24 protein in less than two days from a screen of over 8,000 compounds, with two candidates showing significant in vitro activity [20].

SECTION 07

07 Challenges & Ethical Considerations

7.1 Data Quality & Bias

AI models are only as reliable as the data on which they are trained. Biomedical datasets frequently suffer from systematic biases: clinical trial populations are not demographically representative, published literature over-represents positive results, and proprietary compound activity data is often siloed within pharmaceutical corporations [21]. Models trained on such data may perform admirably on benchmark datasets while failing to generalise to underrepresented disease populations or novel chemical scaffolds — a phenomenon known as distribution shift.

7.2 Interpretability & Regulatory Acceptance

The "black box" nature of deep learning models presents a fundamental challenge for regulatory submission. Regulatory agencies such as the FDA and EMA require mechanistic justification for drug development decisions, yet the internal representations of large neural networks resist straightforward biochemical interpretation [22]. Explainable AI (XAI) techniques — including attention mechanisms, SHAP values, and counterfactual explanations — are being developed to address this gap, but consensus on regulatory standards for AI-assisted submissions remains nascent [22].

> ETHICAL CONSIDERATION
>
> AI-driven drug repurposing may exacerbate healthcare inequities if algorithms are trained predominantly on data from high-income countries, systematically deprioritising diseases prevalent in low- and middle-income settings. Proactive dataset diversification and open-access data sharing are critical countermeasures.

7.3 Intellectual Property & Commercial Incentives

The intellectual property landscape for AI-generated drugs is unresolved. Current legal frameworks in most jurisdictions do not permit non-human inventors to hold patents, creating uncertainty about ownership of AI-designed molecules [23]. Furthermore, the commercial incentives that drive pharmaceutical R&D may not align with repurposing opportunities for off-patent drugs, even when AI identifies them as clinically promising, unless alternative funding mechanisms such as prize models or public-private partnerships are established.

SECTION 08

08 Future Outlook

Several converging technological trends are likely to deepen AI's impact over the next decade. Multimodal foundation models that jointly encode genomic sequences, protein structures, molecular graphs, clinical imaging, and electronic health records will enable more holistic disease modelling than single-modality approaches permit [24]. Foundation models pre-trained on vast biomedical corpora — analogous to GPT-4 in natural language processing — are beginning to emerge, with early examples such as Genie, MolBERT, and GeneFormer demonstrating strong transfer learning performance across diverse downstream drug discovery tasks.

Laboratory automation via self-driving laboratories (SDLs) represents another pivotal development. Platforms such as Emerald Cloud Lab and the Chematica/Synthia route planning system couple AI hypothesis generation with automated synthesis and assay robots, closing the design–make–test–analyse (DMTA) cycle in days rather than months [25]. When combined with generative chemistry, SDLs effectively transform drug discovery into a computational optimisation problem over chemical space.

On the regulatory frontier, the FDA's Advancing Real-World Evidence (RWE) programme and the EMA's PRIME scheme are beginning to accommodate AI-assisted trial design, adaptive protocols, and synthetic control arms — signalling a gradual convergence between regulatory practice and AI-native development workflows [22].

SECTION 09

09 Conclusion

Artificial intelligence is not a panacea for the inherent biological complexity that drives pharmaceutical attrition, but it is a powerful engine for compressing inefficiencies at every stage of the drug development continuum. From AlphaFold's structural revolution to BenevolentAI's COVID-19 repurposing hypothesis and Insilico Medicine's first AI-designed clinical candidate, concrete evidence of transformative impact is accumulating rapidly.

For drug repurposing specifically, AI enables a systematic, data-driven exploitation of approved drug assets against unmet medical needs — an approach with potentially enormous humanitarian and economic returns. Realising this potential will require coordinated investment in open, diverse biomedical data infrastructure; the development of interpretable AI frameworks that satisfy regulatory requirements; and equitable IP arrangements that align commercial incentives with public health priorities.

The pharmaceutical industry is at an inflection point. Organisations that successfully integrate AI into their research culture — not merely as a tool but as a core epistemic methodology — will define the next era of medicine.

References

[1] DiMasi, J. A., Grabowski, H. G., & Hansen, R. W. (2016). Innovation in the pharmaceutical industry: New estimates of R&D costs. *Journal of Health Economics*, 47, 20–33. https://doi.org/10.1016/j.jhealeco.2016.01.012

[2] Schneider, G. (2018). Automating drug discovery. *Nature Reviews Drug Discovery*, 17(2), 97–113. https://doi.org/10.1038/nrd.2017.232

[3] Pushpakom, S., et al. (2019). Drug repurposing: progress, challenges and recommendations. *Nature Reviews Drug Discovery*, 18(1), 41–58. https://doi.org/10.1038/nrd.2018.168

[4] Grand View Research. (2023). *AI in Drug Discovery Market Size Report, 2023–2030*. https://www.grandviewresearch.com/industry-analysis/ai-in-drug-discovery-market-report

[5] Mayr, A., et al. (2018). Large-scale comparison of machine learning methods for drug target prediction on ChEMBL. *Chemical Science*, 9(24), 5441–5451. https://doi.org/10.1039/C8SC00148K

[6] Scannell, J. W., et al. (2012). Diagnosing the decline in pharmaceutical R&D efficiency. *Nature Reviews Drug Discovery*, 11(3), 191–200. https://doi.org/10.1038/nrd3681

[7] Wu, Z., et al. (2018). MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*, 9(2), 513–530. https://doi.org/10.1039/C7SC02664A

[8] Gilmer, J., et al. (2017). Neural message passing for quantum chemistry. *Proceedings of ICML 2017*, 70, 1263–1272. https://arxiv.org/abs/1704.01212

[9] Lin, Z., et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637), 1123–1130. https://doi.org/10.1126/science.ade2574

[10] Jumper, J., et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

[11] Tunyasuvunakool, K., et al. (2021). Highly accurate protein structure prediction for the human proteome. *Nature*, 596(7873), 590–596. https://doi.org/10.1038/s41586-021-03828-1

[12] Ren, F., et al. (2023). AlphaFold accelerates artificial intelligence powered drug discovery: efficient discovery of a novel CDK20 small molecule inhibitor. *Chemical Science*, 14, 1443–1452. https://doi.org/10.1039/D2SC05709C

[13] Zhavoronkov, A., et al. (2019). Deep learning enables rapid identification of potent DDR1 kinase inhibitors. *Nature Biotechnology*, 37(9), 1038–1040. https://doi.org/10.1038/s41587-019-0224-x

[14] Cheng, F., et al. (2018). Network-based approach to prediction and population-based validation of in silico drug repurposing. *Nature Communications*, 9, 2691. https://doi.org/10.1038/s41467-018-05116-5

[15] Sang, H., et al. (2019). Predicting drug–disease associations through layer attention graph convolutional network. *Briefings in Bioinformatics*, 22(4). https://doi.org/10.1093/bib/bbaa243

[16] Lee, J., et al. (2020). BioBERT: a pre-trained *Bioinformatics*, 36(4), 1234–1240.

biomedical language representation model for biomedical text mining. https://doi.org/10.1093/bioinformatics/btz682

[17] Richardson, P., et al. (2020). Baricitinib as potential treatment for 2019-nCoV acute respiratory disease. *The Lancet*, 395(10223), e30–e31. https://doi.org/10.1016/S0140-6736(20)30304-4

[18] Beck, B. R., et al. (2020). Predicting commercially available antiviral drugs that may act on the novel coronavirus (SARS-CoV-2) through a drug-target interaction deep learning model. *Computational and Structural Biotechnology Journal*, 18, 784–790. https://doi.org/10.1016/j.csbj.2020.03.025

[19] Exscientia. (2020). *Exscientia and Sumitomo Dainippon Pharma advance world's first AI-designed drug into clinical trials*. Press Release. https://www.exscientia.ai/news-insights/dsp-1181

[20] Wallach, I., et al. (2015). AtomNet: A Deep Convolutional Neural Network for Bioactivity Prediction in Structure-Based Drug Discovery. *arXiv*. https://arxiv.org/abs/1510.02855

[21] Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future — big data, machine learning, and clinical medicine. *New England Journal of Medicine*, 375(13), 1216–1219. https://doi.org/10.1056/NEJMp1606181

[22] FDA. (2021). *Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) Action Plan*. U.S. Food and Drug Administration. https://www.fda.gov/media/145022/download

[23] Abbott, R., & Storz, A. (2019). The Artificial Inventor. *AIPLA Quarterly Journal*, 46(3). https://ssrn.com/abstract=3359756

[24] Bommasani, R., et al. (2021). On the opportunities and risks of foundation models. *arXiv*. https://arxiv.org/abs/2108.07258

[25] Coley, C. W., et al. (2019). A robotic platform for flow synthesis of organic compounds informed by AI planning. *Science*, 365(6453). https://doi.org/10.1126/science.aax1566